Numerical Computing II — Assignment V: Lagrange Interpolation on Real Life data

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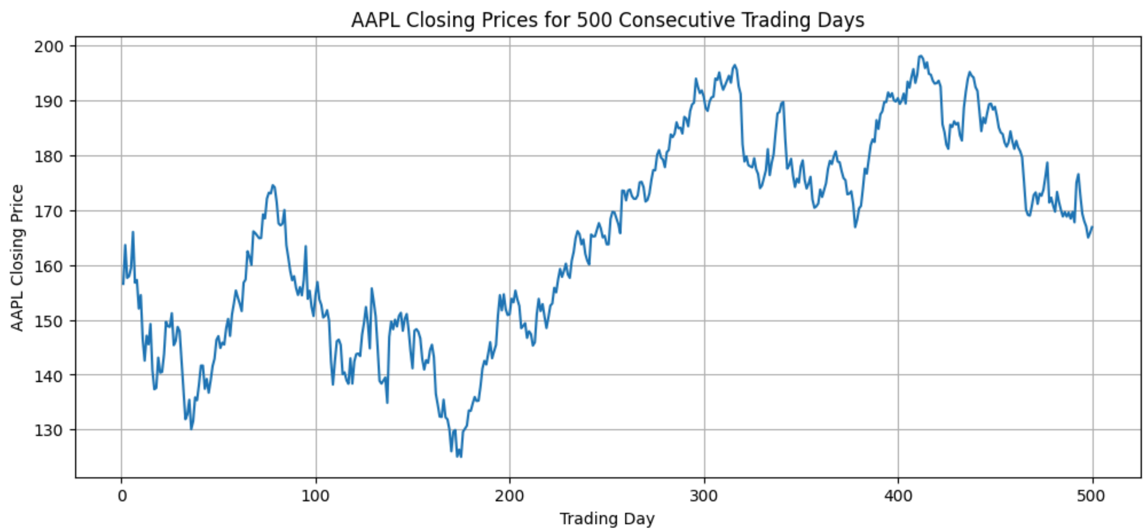
1. Dataset Overview and Context

For this experiment, historical stock-price data of Apple Inc. (AAPL) are used to investigate the behavior of Lagrange interpolation on a large real-world dataset. The dataset contains daily stock-market information such as the opening price, highest price, lowest price, closing price, adjusted closing price, and trading volume.

The complete dataset contains 10,931 observations (shape: 10,931 × 7 with no missing values). For the interpolation experiment, 500 consecutive observations are selected from the end of the dataset, corresponding to the period from 27 April 2022 to 23 April 2024. The Close column is used as the dependent variable because the objective is to study daily closing stock prices.

Table 1 — Dataset information

Property	Value
Dataset	AAPL historical stock prices
	https://www.kaggle.com/datasets/hchsmost/test-dataset
Total observations	10,931
Observations used	500
Selected period	27-Apr-2022 to 23-Apr-2024
Variable interpolated	Closing price
Independent variable	Trading-day index (1 to 500)



The plot shows variation in the daily closing price; this is a real-world dataset for investigating the limitations of global polynomial interpolation.

2. Lagrange Implementation

The Lagrange interpolation polynomial is implemented programmatically using the following function:

```
def numerical_lagrange(x_data, y_data, x_val):
    n = len(x_data)
    total_p = 0.0
    for i in range(n):
        li = 1.0
        for j in range(n):
            if j != i:
                li *= (x_val - x_data[j]) / (x_data[i] - x_data[j])
        total_p += y_data[i] * li
    return total_p
```

At an interpolation node x_i, the Lagrange polynomial should reproduce the corresponding observed value y_i:

```
x = 1, actual = 156.570007, interpolated = 156.570007, absolute error =
0.000e+00
x = 2, actual = 163.639999, interpolated = 163.639999, absolute error =
0.000e+00
x = 250, actual = 163.770004, interpolated = 163.770004, absolute error =
0.000e+00
x = 500, actual = 166.899994, interpolated = 166.899994, absolute error =
0.000e+00
```

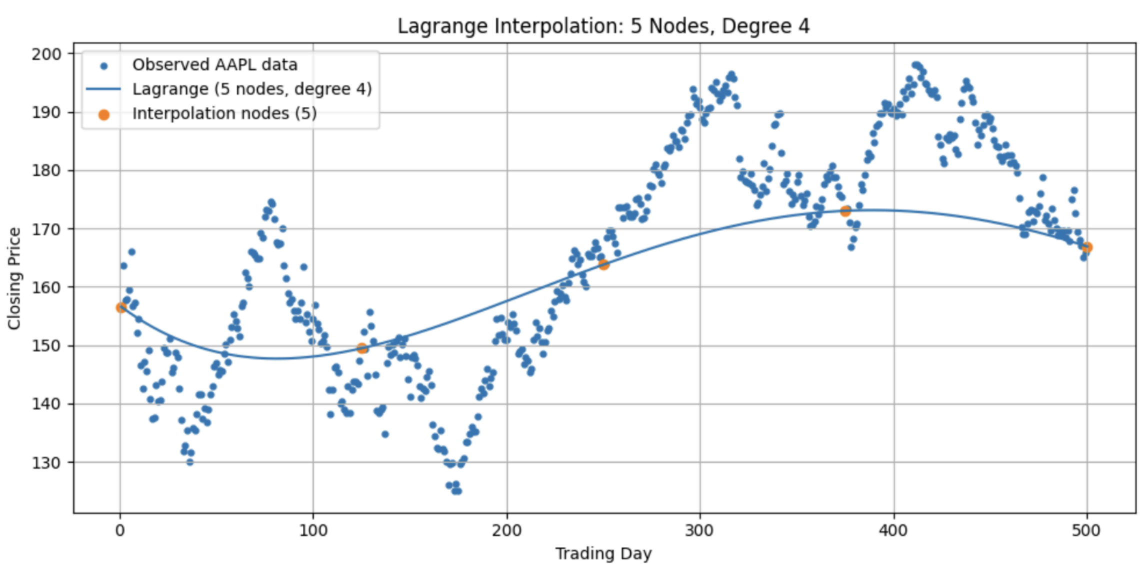
3. Increasing Number of Interpolation Nodes

To investigate the effect of polynomial degree, different numbers of equally spaced interpolation nodes were selected from the same 500 observations using: n = 5, 10, 20, 50, 100, 250, 500.

Table 2 — Number of nodes and corresponding polynomial degrees

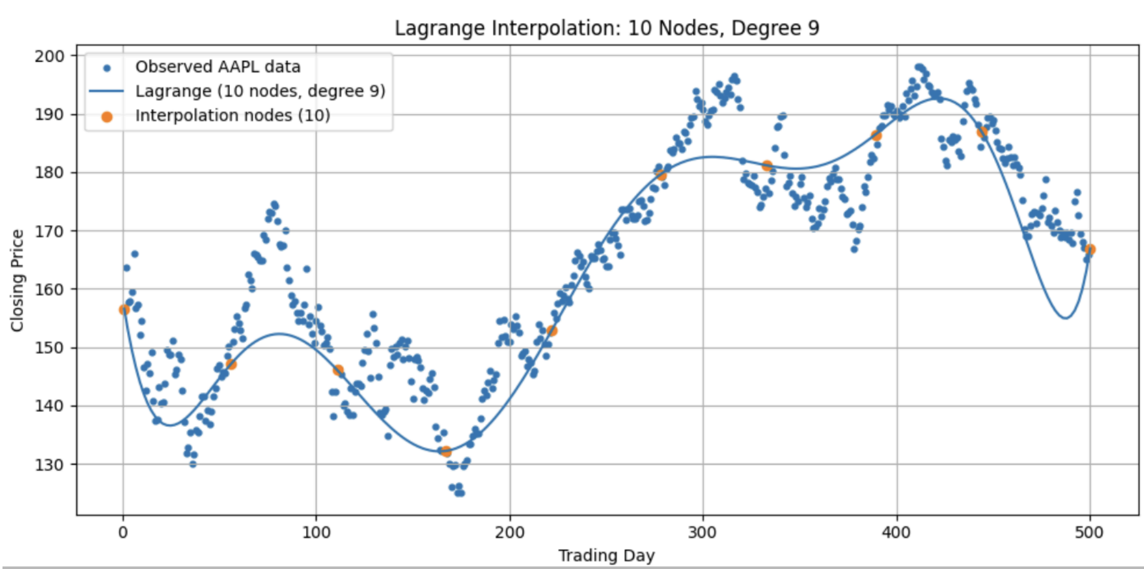
Number of Nodes (n)	Polynomial Degree (n - 1)
5	4
10	9
20	19
50	49
100	99
250	249
500	499

Figure 1: 5 Nodes — Degree 4



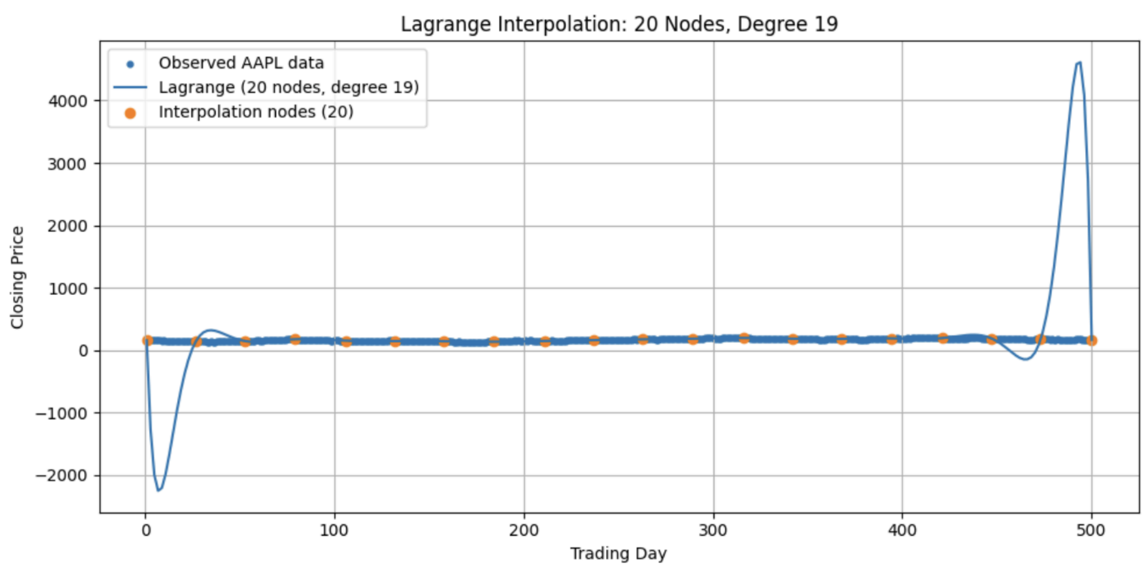
The polynomial is smooth. It does not reproduce the fluctuations because only five observations were used.

Figure 2: 10 Nodes — Degree 9



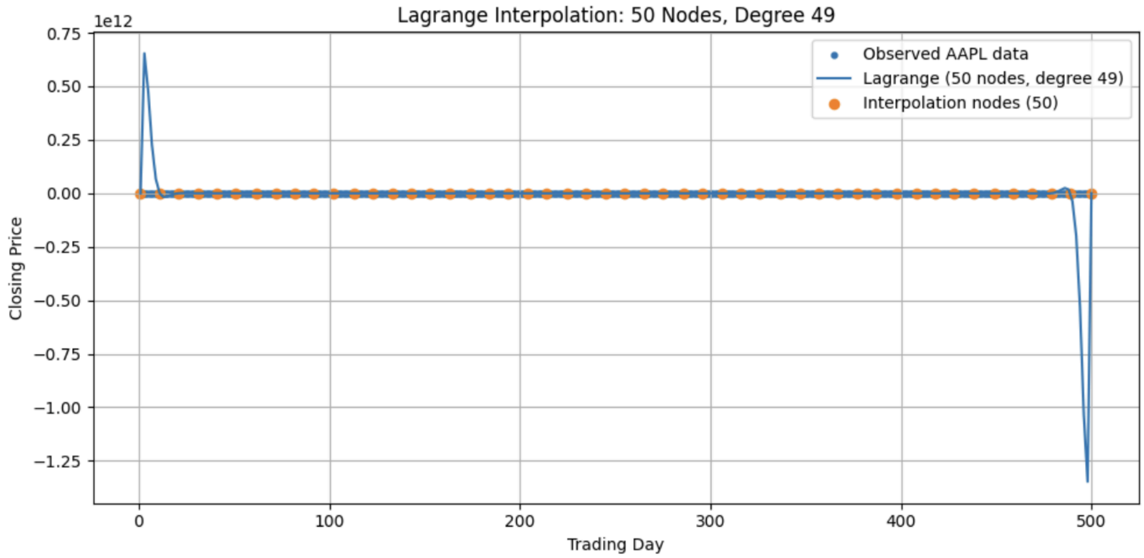
The polynomial follows the overall trend more closely.

Figure 3: 20 Nodes — Degree 19



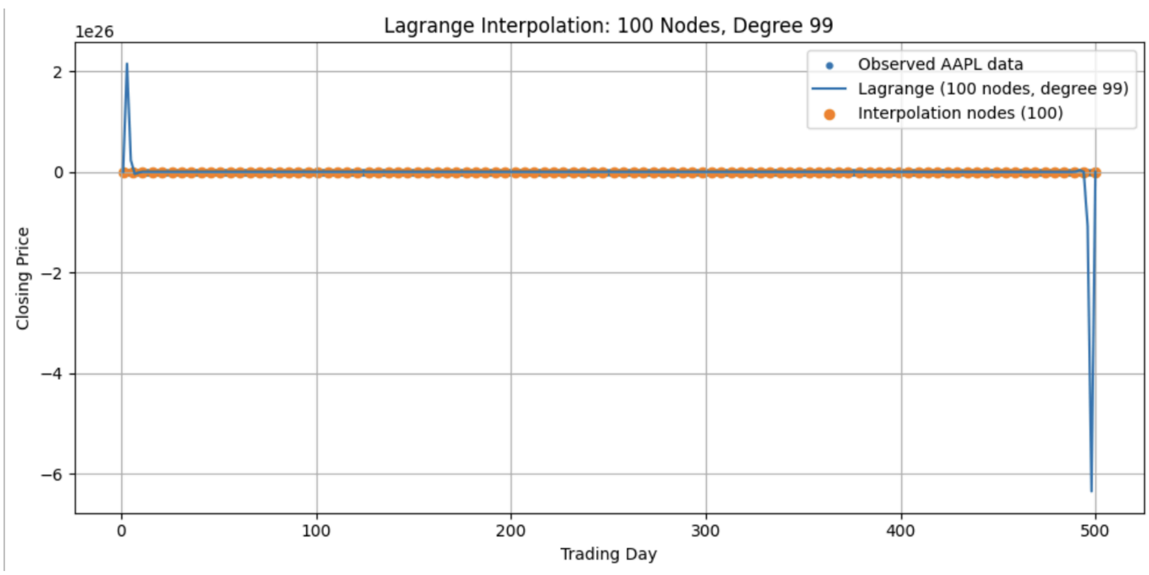
Large oscillations begin to appear, particularly near the boundaries of the interval.

Figure 4: 50 Nodes — Degree 49



The oscillations become extremely large. The polynomial begins producing values many orders of magnitude larger than the actual stock prices.

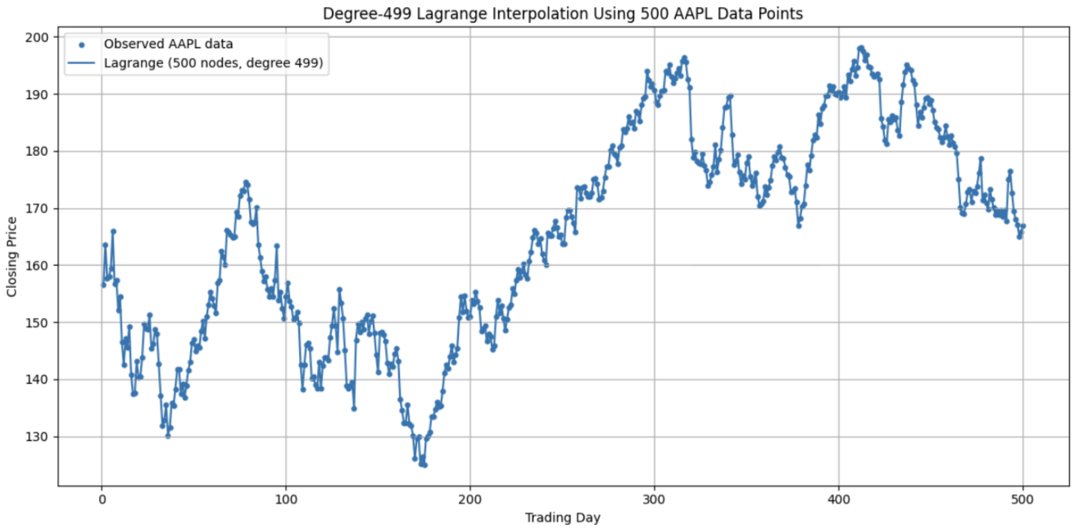
Figure 5: 100 Nodes — Degree 99



The numerical instability becomes huge. The interpolant is no longer a meaningful representation of the stock-price series between the nodes.

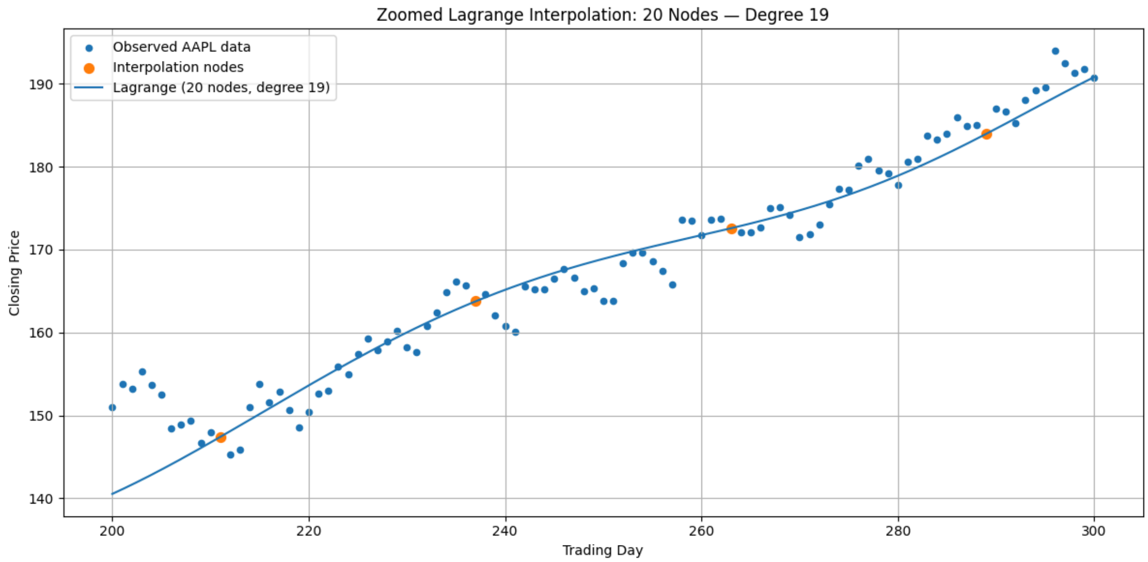
Increasing the number of interpolation nodes initially improves the ability of the polynomial to follow the behavior of the data. But the polynomial degree becomes large and the data develops oscillations and instability.

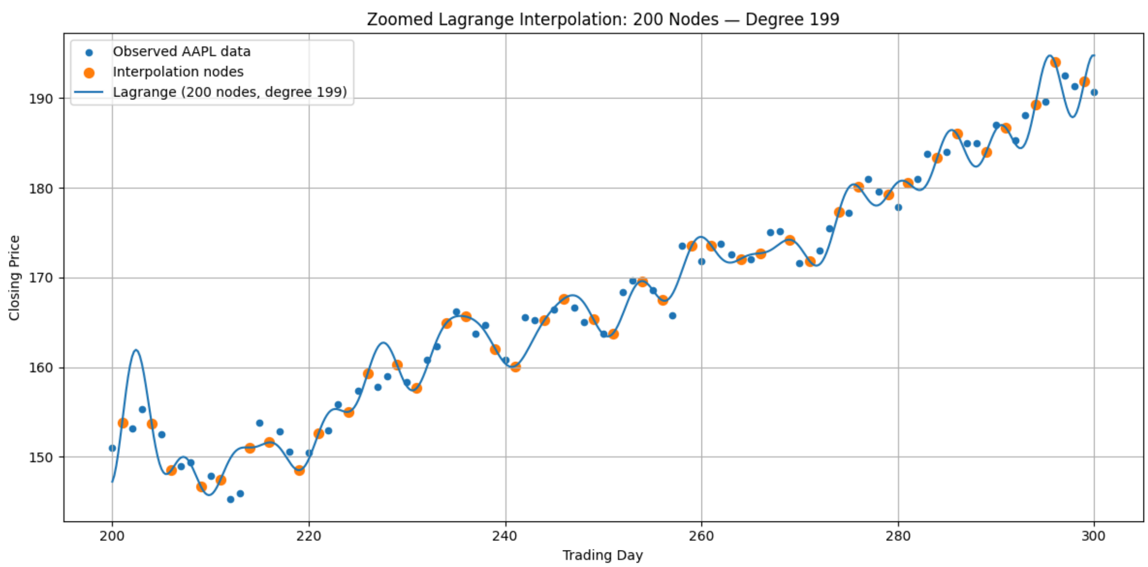
Figure 6: 500 Nodes — Degree 499



The 500-node interpolation. Standard Interpolation.

Figure 7: Zoomed-In Analysis





4. Conclusion

In this experiment, numerical Lagrange interpolation was applied to 500 consecutive AAPL closing-price observations.

The effect of increasing the number of interpolation nodes was studied using 5, 10, 20, 50, 100, 250, and 500 nodes, corresponding to polynomial degrees from 4 to 499.

The 500-node experiment demonstrated an important limitation of global polynomial interpolation: satisfying all interpolation conditions does not guarantee a useful approximation between the nodes. Thse high-degree polynomial can produce values many orders of magnitude larger than the actual stock-price data.